

Technology #08

NIR Photovoltaic Power Delivery for
Implantable Devices

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

NIR Photovoltaic Power Delivery for Implantable Devices

WHY DOES IT MATTER?

Implantable sensors need power without batteries or leads

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 9 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Optics + Photonics + Power Electronics. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Implantable device company; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

940nm optical power delivery concept + verified fluence (T2-CONFIRMED); Computational model (PyTissueOptics v2.0.1, DCC-Lab independent verification); Optical properties specification (Jacques 2013); Bench experiment protocol (LED + phantom + PV); External precedent: PMC5646820 confirms subcutaneous PV IR harvesting; This engineering dossier with optical-specific design chain

WHAT DOES THE BUYER HAVE TO BUILD?

Implantable PV cell packaging (biocompatible, optically transparent at 940nm); Thermal management design (LED heating + PV heating); External LED patch design (wearable, chronic use); Power conditioning circuit (implantable); Regulatory submission (PMA — optical implant); Clinical validation evidence

WHAT IS THE NEXT DECISIVE EXPERIMENT?

NIR source + PV cell bench test

WHAT WOULD MAKE US KILL IT?

Tissue attenuation at 5cm depth prevents sufficient power delivery ($>1\text{mW/cm}^2$ required at PV surface)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: NIR source + PV cell bench test

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - NIR source + PV cell bench test - 10 weeks

Basis: Build plan WP. Scope: P_electrical vs P_optical at various depths. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - PV cell at low irradiance - 6 weeks

Basis: Build plan WP. Scope: PV efficiency vs irradiance. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Hermetic encapsulation of PV cell - 10 weeks

Basis: Build plan WP. Scope: Hermeticity per MIL-STD-883 + optical transparency. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: NIR window (~800-1000 nm) provides optimal tissue penetration; PV cell efficiency retained at low irradiance; External source can deliver sufficient optical power without tissue heating; PV cell biocompatible when encapsulated. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: NIR source + PV cell bench test. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: External 940nm NIR source penetrates tissue; implanted GaAs PV cell converts to electrical energy

POTENTIAL VALUE DRIVER: Addresses: Implantable sensors need power without batteries or leads

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Implantable device company; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Optics + Photonics + Power Electronics

Disciplines: Optical engineering, Photovoltaic engineering, Tissue optics, Thermal management, Implantable electronics

Architecture: External 940nm LED → scalp/skull tissue → implanted GaAs PV cell → power conditioning → sensor/load

Subsystems:

SS-01: External 940nm LED - Generate collimated NIR light at 940nm

SS-02: Tissue transmission path - Optical transmission through scalp+skull (~5mm)

SS-03: GaAs PV cell - Convert 940nm photons to electrical power

SS-04: Power conditioning - Regulate PV output for sensor/load

2. Mechanism Architecture

Physical Changes: NOT ESTABLISHED

Key Physics: NOT ESTABLISHED

3. Governing Engineering Model

Summary: Near-infrared photovoltaic power delivery through tissue for implantable shunt sensor. NIR light from external source penetrates tissue; implanted PV cell converts to electrical energy.

Equations:

Beer-Lambert: $I(d) = I_0 * \exp(-\mu_{eff} * d)$ [P-16 NIR tissue attenuation through scalp/skull to implanted PV cell]

$P_{electrical} = P_{optical} * \eta_{PV}$ [P-16 PV conversion at implanted catheter PV cell]

$\mu_{eff} \sim 1-10 \text{ cm}^{-1}$ in tissue at 800-1000 nm (EXTERNAL_PRECEDENT)

Assumptions:

- NIR window (~800-1000 nm) provides optimal tissue penetration
- PV cell efficiency retained at low irradiance
- External source can deliver sufficient optical power without tissue heating
- PV cell biocompatible when encapsulated

Boundary Conditions:

- Wavelength: 800-1000 nm NIR window
- Tissue depth: 1-5 cm typical for shunt placement
- PV cell size: constrained by catheter geometry

Failure Regimes:

- Tissue depth too large -> $P_{optical}$ insufficient
- PV efficiency too low at low irradiance
- Tissue heating from external source exceeds safety limit
- PV cell degradation in implant environment

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Transcranial power for implanted devices. Current solution (inductive coupling) is alignment-critical and depth-limited.	VERIFIED
DI-002	Functional requirement	940nm NIR through scalp/skull to implanted GaAs PV cell. Verified fluence 100-1400 $\mu\text{J}/\text{cm}^2$ (~1000-1400 μW).	MODELLED
DI-003	Power output target	$\geq 500 \mu\text{W}$ (pass threshold)	MODELLED
DI-004	Wavelength	940nm	MODELLED
DI-005	PV cell type	GaAs (Gallium Arsenide)	MODELLED
DI-006	Tissue path length	~5mm (scalp + skull)	MODELLED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Thermal safety	UNKNOWN	UNKNOWN
DI-009	Long-term LED/skin interface stability	UNKNOWN	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	940nm LED specifications	CONCEPTUAL	
DO-002	GaAs PV cell specifications	CONCEPTUAL	
DO-003	Tissue phantom (5mm)	DEFINED	
DO-004	Implant packaging	ABSENT	PV cell area; Implant geometry; Encapsulation material; Thermal management

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
P-16 Wavelength	EXTERNAL_PRECEDENT (~800-1000 nm NIR window)	nm	Published tissue optical properties	
Tissue attenuation coefficient	EXTERNAL_PRECEDENT (~1-10 cm ⁻¹ in NIR window)	cm ⁻¹	Published tissue optics	
PV efficiency at low irradiance	UNKNOWN for targeted PV cell %	%	PV cell selection	
External source power	UNKNOWN — bounded by tissue heating limit	W	Safety constraint	
PV cell area	UNKNOWN — constrained by catheter geometry	cm ²	Catheter integration	
Tissue depth	EXTERNAL_PRECEDENT (1-5 cm typical)	cm	Anatomical constraint	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Insufficient power at depth	Tissue attenuation > link budget	Higher source power (limited by safety)	Ray safety shoe measurements	Longitudinal tissue depth
Tissue heating from external source	External source heats tissue	Power limit + duty cycling + cooling	Testing heating test per IEC 60601-1-2	UNKNOWN achievable power within safety limit
PV efficiency drops at low irradiance	PV cells typically less efficient at low intensity	Low-light optimized PV cells	PV cell characterization at low intensity	UNKNOWN achievable efficiency
PV cell degradation in long-term use	Long-term degradation in CSF tissue	Stable encapsulation + derating	Aging study at 37°C in CSF	UNKNOWN degradation rate
Source-catheter alignment error	External source not aligned with implanted PV cells	Large Planar PV cells source alignment tolerance	Alignment tolerance test	UNKNOWN achievable alignment tolerance clinically
Biocompatibility of PV materials	PV materials (semiconductors, metals, encapsulation)	Use biocompatible materials (e.g. IS-1093 Series)	ISO 10993 Series	UNKNOWN chronic biocompatibility

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Insufficient power at depth	Tissue attenuation > link budget	Higher source power (safety-limited); slanted waveguide, larger PV	UNKNOWN achievable power at 5 cm
Tissue heating from external source	External source heats tissue	Power limit + duty cycling + cooling	UNKNOWN achievable power within safety limit
PV efficiency at low irradiance	PV less efficient at low intensity	Low-irradiance-optimized PV cell	UNKNOWN achievable efficiency
PV cell degradation in implant	Long-term degradation in CSF/tissue	Hermetic encapsulation + derating	UNKNOWN degradation rate
Source-catheter alignment	External source not aligned with PV	Larger PV + diffuse source + alignment	UNKNOWN achievable alignment clinically
Biocompatibility of PV materials	PV materials not biocompatible	Hermetic encapsulation	UNKNOWN chronic biocompatibility

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Measured power >= 500 μ W	Bench: 940nm LED + 5mm tissue phantom	500 μ W, 940nm PV cell	NOT_TESTED
V-002	Fluence verification (T2)	PyTissueOptics v2.0.1 Monte Carlo (independent code base)	100% coverage of published range 30-400 mW/cm ² , CIs overlap, Jacques	PASSED
V-003	Thermal safety	Thermal simulation + bench measurement	Temperature rise < 2°C at tissue	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Sufficient power delivery for chronic implant	Portable system operation end-to-end power delivery in patients	Pre-delivery in patients, on power	NOT_TESTED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
PV cell	GaAs (Gallium Arsenide)	R332 mechanism explicitly specifies GaAs	ISO 10993 biocompatibility for G332	SOURCE: ENEC + not a problem
PV encapsulation	UNKNOWN	No encapsulation material identified	Material selection + ISO 10993	Upcoming transparency at 940nm

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	940nm LED	1	COTS_COMPONENT	Semiconductor (GaAs)	Multiple commercial suppliers available	High	Optical power output + thermal test
02	GaAs PV cell	1	COTS_COMPONENT	GaAs	Multiple commercial suppliers available	Medium	Power conversion efficiency + biocompatibility
03	Tissue phantom (5mm)	1	TEST_FIXTURE	Tissue-mimicking polymer with acoustic properties	Customized by research lab	Medium	Optical property verification

13. Manufacturing

Status: ENGINEERING_CANDIDATE — implant packaging for GaAs PV in CSF environment not defined

Candidate Processes:

- PV cell dicing + packaging: Standard semiconductor packaging -
- LED module assembly: Standard LED packaging -

14. External Evidence

Source 1: Subcutaneous Photovoltaic Infrared Energy Harvesting for Bio - PMC

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5646820>

Excerpt: The PV cells may be stacked on the implantable system, and include an integrated photodiode to provide a means of low-power wireless optical communication to ...

Source 2: [PDF] Subcutaneous Photovoltaic Infrared Energy Harvesting for Bio

URL: <https://www.semanticscholar.org/paper/Subcutaneous-Photovoltaic-Infrared-Energy-for-Moon-Blaauw/18fae96f6f7351833fa96dd7f5e22af54375394d>

Excerpt: This work shows that photovoltaic cells at the millimeter-scale can achieve a power conversion efficiency of more than 17% for silicon and 31% for GaAs ...

Source 3: GaAs-Based Photovoltaic Infrared Energy Harvesting for Microscale

URL: https://www.researchgate.net/publication/365383611_GaAs-Based_Photovoltaic_Infrared_Energy_Harvesting_for_Microscale_Biomedical_Implants

Excerpt: GaAs-Based Photovoltaic Infrared Energy. A method for efficient, rapid, and minimally invasive implantation ■ utilizes custom-designed PV cells specifically ...

Source 4: Near-Infrared Subcutaneous Generator for Powering Implantable

URL: <https://journal.sitp.ac.cn/hw/hwen/article/html/H2026025>

Excerpt: This article reviews the latest advances in light transmission enhancement strategies and energy conversion mechanisms in near-infrared subcutaneous generators.

Source 5: [PDF] Dual-Junction GaAs Photovoltaics for Low Irradiance Wireless Power

URL: <https://blaauw.engin.umich.edu/wp-content/uploads/sites/342/2021/02/Dual-Junction-GaAs-Photovoltaics-for-Low-Irradiance-Wireless-Power-Transfer-in-Submillimeter-Scale-Sensor-Nodes.pdf>

Excerpt: Feb 3, 2021 · Near-infrared (NIR) light provides a means for high-power conversion efficiency (>30%) in millimeter-scale subcutaneous implantable devices [14] ...

15. Transfer Boundary

YOU RECEIVE:

- + 940nm optical power delivery concept + verified fluence (T2-CONFIRMED)
- + Computational model (PyTissueOptics v2.0.1, DCC-Lab independent verification)
- + Optical properties specification (Jacques 2013)
- + Bench experiment protocol (LED + phantom + PV)
- + External precedent: PMC5646820 confirms subcutaneous PV IR harvesting
- + This engineering dossier with optical-specific design chain

YOU MUST DEVELOP:

- > Implantable PV cell packaging (biocompatible, optically transparent at 940nm)
- > Thermal management design (LED heating + PV heating)
- > External LED patch design (wearable, chronic use)
- > Power conditioning circuit (implantable)
- > Regulatory submission (PMA — optical implant)
- > Clinical validation evidence

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable power at target tissue depth (5 cm) — UNKNOWN (critical for feasibility)
- PV efficiency at low irradiance — UNKNOWN
- External source power limit given tissue heating — UNKNOWN
- PV cell degradation rate in implant environment — UNKNOWN
- Source-catheter alignment tolerance clinically — UNKNOWN
- Regulatory pathway (active implantable, optical safety) — UNKNOWN
- Clinical utility (sufficient power for what sensor/telemetry?) — UNKNOWN

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	NIR source + PV cell	LED, PV cell, optical power meter	Optical power at target depth	Optical power at target depth	Optical power at target depth	10 weeks
WP-02	Tissue heating test article	Tissue phantom, thermocouple	Temperature rise with LED	Temperature rise with LED	Temperature rise with LED	6 weeks
WP-03	PV cell at low irradiance	Solar simulator, IV characterizer	Efficiency vs irradiance	Efficiency meets target	Efficiency vs irradiance	1 week
WP-04	PV cell aging specimen	Aging chamber at 37°C	Efficiency vs time	Efficiency retained > 80%	Efficiency vs time	52 weeks (chronic)
WP-05	Hermetic encapsulation	Profilometer or Ti characterizing	Hermeticity	Hermeticity	Hermeticity	4 weeks
WP-06	Integrated catheter-PV	Prototype + electrical test	End power delivery	Sensor operates from integrated power	Integrated power	12 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Implantable device company; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Tissue attenuation at 5cm depth prevents sufficient power delivery ($>1\text{mW}/\text{cm}^2$ required at PV surface)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Tissue attenuation at 5cm depth prevents sufficient power delivery ($>1\text{mW}/\text{cm}^2$ required at PV surface)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.